

Weights are positive whole amounts. A card cannot be split or used twice. The target stays on the left pan. Put a weight on either pan or leave it off.

Problem 1: Use the kit 1, 2, 3. Your partner removes one weight. Balance each target 1, 2, 3 with the weights that remain. Find balances for every possible removal.

1	2	3
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target + weights	=	weights
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Removed	Target	Left-pan weights	Right-pan weights
1	1		
1	2		
1	3		
2	1		
2	2		
2	3		
3	1		
3	2		
3	3		

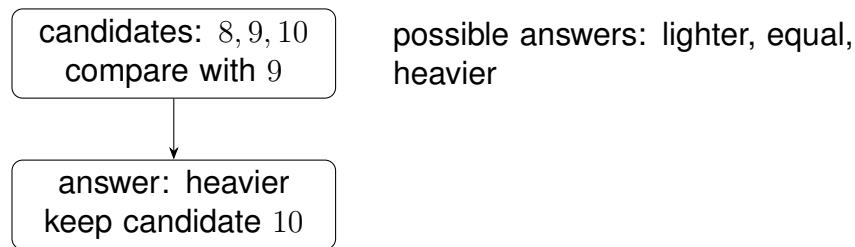
Problem 2: Choose three different weights from 1 to 8. Your partner may remove any one of them before choosing a target. Make a kit that can still balance every target from 1 to 3. Can a different kit do this?

1	2	3	4	5	6	7	8
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Problem 3: Can three different whole-number weights still balance every target from 1 to 4 after any one weight is removed? Choose weights of any size. Find a kit or explain why no kit can work.

Compare a hidden mass with any known whole-number mass. Your partner answers *lighter*, *equal*, or *heavier*. Guessing the only remaining candidate uses no comparison.



Problem 4: Your partner hides one card from 1 to 7. Make a plan that always finds it in at most two comparisons. Test your plan with your partner using each possible hidden card.

1	2	3	4	5	6	7
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Problem 5: Could a plan always find a hidden card from 1 to 8 in at most two comparisons? Explain your answer.

For teams, use every listed weight once. Each team needs at least one weight. All team totals must match.

Problem 6: Put each kit into the requested number of equal-weight teams. If a kit cannot be divided that way, explain why.

Kit 1, 2, 3, 4
2 teams

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Kit 1, 2, 3, 4, 5, 6
3 teams

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Kit 1, 2, 5
2 teams

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Kit 1, 2, 3, 4, 5, 6, 7, 8
4 teams

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Problem 7: Find every way to split 1, 2, 3, 4, 5, 6 into three equal-weight teams. Moving a whole team to another tray does not make a new way.